Supplementary Materials

Transfer learning from computed stability data for nanobody melting-temperature prediction

Taihei Murakami, Kentaro Sasaki, Soichiro Oda, Kazuma Okada, and Yasuhiro Matsunaga

Supplementary methods

Details of the heterogeneous MD data set

The heterogeneous data set was assembled from a SAbDab summary and the corresponding structures [1]. For each PDB identifier, we selected the first heavy-chain entry whose chain was present in the structure file. After requiring a usable sequence, topology, trajectory, and hydrophilic native-contact label, 763 samples remained, each a distinct amino-acid sequence. The model did not receive the PDB identifier, and sequences exceeding the 158-residue model input were truncated at the end.

Structures were prepared and protonated at pH 7.4 with PDBFixer, pdb2pqr, and PROPKA [2, 3]. Each protein was solvated in OPC water with 15 Å padding and 0.15 M NaCl and described with ff19SB [4,5]. Equilibration at 300 K used 1 ns NVT and 1 ns NPT with a 2-fs timestep, after which two 1-ns restrained NVT stages at 400 K preceded 100 ns of unrestrained 400-K NVT production. Production used hydrogen-mass repartitioning to 4 amu, a 4-fs timestep, and output every 100 ps. OpenMM used PME electrostatics with a 1.0-nm cutoff, constraints on bonds to hydrogen, and a Langevin middle integrator with 1 ps^{-1} friction [6].

The contact calculation used backbone heavy-atom pairs separated by more than three residues and within 0.45 nm in the reference structure, counting a contact only when at least one residue was hydrophilic, as for the single mutation scans. Each structure was referenced to its own parent equilibration structure, and Q was averaged over the $[10, 40)$ ns production window sampled every 100 ps (300 frames). The two MD data sets thus shared one Q definition, contact set, native reference, and averaging window, and differed only in the sequences they covered.

FEP charge-change check

For charge-changing mutations in a cubic 70-Å box and a solvent dielectric constant of 78, we tested the leading analytical periodic net-charge (Ewald self-energy) correction [7], $0.086(\Delta q)^2$ kcal mol^{-1}. We applied it and repeated the sign reversal, robust scaling, Yeo–Johnson transform with standardization, and min–max scaling used for the training labels. Across the 284 charge-changing mutations the scaled labels barely moved, with a maximum shift of 0.008 on the $[0, 1]$ scale and a correlation of 0.99989 with the uncorrected labels, so the training labels use the uncorrected $\Delta\Delta G$.

Model-size checks

We fine-tuned the 35M- and 650M-parameter ESM2 encoders without size-specific hyperparameter searches. For both sizes, the Tm-only model used the shared architecture, dropout 0.05, and an encoder learning rate of 1×10^{-4}. The FEP model used the shared architecture with separate 1MEL and 4IDL heads, $n = 320$ samples per structure, dropout 0.15, and an encoder learning rate of 3×10^{-5}. Both conditions used a non-encoder learning rate of 3×10^{-4}, weight decay 0.04, and ten training seeds.

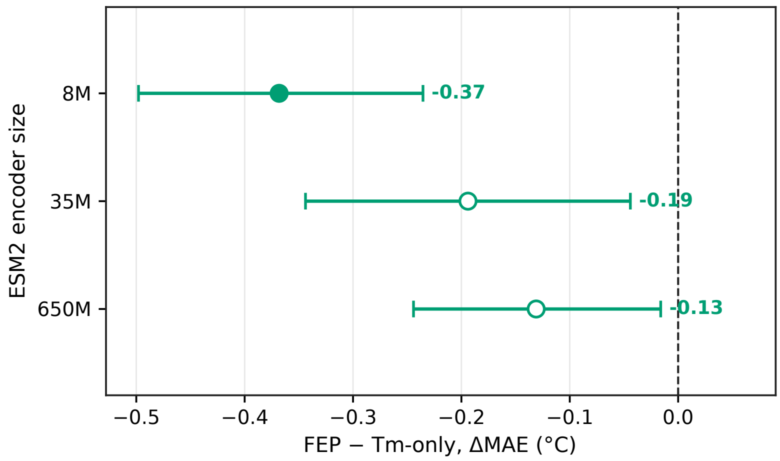

Figure S1 The FEP result held across ESM2 encoder sizes. FEP-minus-Tm-only MAE (ΔMAE) with a fine-tuned encoder; negative values mean lower Tm error with FEP. The 8M point is the 24-model ensemble used in the main physical-observable comparison, and the 35M and 650M points are ten-seed checks without size-specific tuning. Whiskers are 95% paired bootstrap intervals over the 396 Tm test sequences.

References

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